

MRMU Example: Urate to CAD_UKB With LDSC Parameters and Local PLINK Clumping

Generated with MRMU R package

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Summary

This report reruns the urate-CAD example using both newly supplied production resources:

- 1KG EUR PLINK bfile for local LD clumping
- EUR LDSC scores for estimating Ω , C, and using real SNP-level L2

Analysis setup:

- Exposure: Biomarker_Urate
- Outcome: CAD_UKB
- Confounders: Metabolic_SBP, Metabolic_DBP
- IV selection threshold: 5e-5
- Clumping: local PLINK 1.9, clump_kb = 1000, clump_r2 = 0.001
- LDSC: pairwise estimation across Biomarker_Urate, Metabolic_SBP, Metabolic_DBP, and CAD_UKB

exposure	outcome	confounder	beta	SE	p-value	clumped IVs	effective IVs	pi0
Biomarker_Urate	CAD_UKB	Metabolic_SBP, Metabolic_DBP	0.015260	0.41919	0.41919	696	322.231	0.462975

Data

The analysis used the organised files under:

path/to/MRMU_R_github

folder	role
raw_gwas/	GWAS summary statistics for urate, SBP, DBP, and CAD_UKB

folder	role
plink_reference/ ldscore/eur_w_ld_chr/	all_1000G_EUR_Phase3.bed/.bim/.fam LDSC score files 1-22.12.ldscore.gz and 1-22.12.M_5_50
tools/plink_mac/	macOS PLINK 1.9 executable

After harmonising alleles across the four traits with real LDSC L2, the merged data contained 1,012,146 SNPs.

Local Clumping

PLINK log summary:

item	value
candidate urate IVs at $p \leq 5e-5$	14,090
LD clumps formed	696
removed due to LD/reference absence	13,394

The difference from the previous clumped fallback run is expected: this run uses the real LDSC-merged SNP set, while the earlier fallback used $L2 = 1$.

LDSC Parameters

Pairwise LDSC results:

trait1	trait2	rg	omega	C
Biomarker_Urate	Biomarker_Urate	1.0000	1.163e-07	1.1123
Biomarker_Urate	Metabolic_SBP	0.1553	1.634e-08	0.0579
Biomarker_Urate	Metabolic_DBP	0.1316	1.331e-08	0.0666
Biomarker_Urate	CAD_UKB	0.2735	2.238e-08	0.0529
Metabolic_SBP	Metabolic_SBP	1.0000	9.822e-08	1.2026
Metabolic_SBP	Metabolic_DBP	0.7748	7.293e-08	0.8589
Metabolic_SBP	CAD_UKB	0.3625	2.747e-08	0.0460
Metabolic_DBP	Metabolic_DBP	1.0000	9.110e-08	1.1833
Metabolic_DBP	CAD_UKB	0.2798	2.044e-08	0.0163
CAD_UKB	CAD_UKB	1.0000	5.970e-08	1.0296

MR-MU Result

quantity	value
beta	0.01232753
standard error	0.0152602
p-value	0.41919
clumped IVs	696
effective IVs	322.2309
pi0	0.4629754
tau.sq	2.368078e-07
threshold	5e-5
method label	MR-MU(LDSC Omega/C, local PLINK clump)

Interpretation

After adding real LDSC parameters (**Omega**, **C**) and real SNP-level L2, the estimated urate effect on CAD_UKB is small and not statistically significant in this MR-MU run.

This LDSC-based result supersedes the earlier fallback reports because it uses the LD score data and the supplied local PLINK reference.

Reproducibility

The example was run from the package root:

```
MRMU_EXAMPLE_DIR=/path/to/MRMU_R_github Rscript scripts/run_urate_cad_mrmu_ldsc_example.R
```

Key output files:

file	description
results/urate_cad_mrmu_ldsc_clumped/LDSC-based-MRMU-ldsc-result.csv	LDSC-based MRMU-ldsc result
results/urate_cad_mrmu_ldsc_clumped/urate_cad_mrmu_ldsc_fit.rds	urate_cad_mrmu_ldsc_fit
results/urate_cad_mrmu_ldsc_clumped/urate_cad_mrmu_ldsc_output.rds	urate_cad_mrmu_ldsc_output
results/urate_cad_mrmu_ldsc_clumped/MRMU-ldsc-gene-urates_sbp_dbp_cadukb.rds	MRMU-ldsc-gene-urates_sbp_dbp_cadukb
results/urate_cad_mrmu_ldsc_clumped/p-value-palldsc-results.csv	p-value-palldsc-results
results/urate_cad_mrmu_ldsc_clumped/runlog	runlog